

TOMMOT, Russian Federation

WMO: 310050

Lat: 58.967N

Lon: 126.267E

Elev: 284

StdP: 97.96

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -49.0	(c) -47.0	(d) -50.7	(e) 0.0	(f) -48.9	(g) -48.8	(h) 0.0	(i) -47.0	(j) 4.2	(k) -17.9	(l) 3.4	(m) -20.5	(n) 0.2	(o) 180	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	29.7	18.2	27.4	17.4	25.4	16.6	20.2	26.3	19.0	25.3	17.9	23.4	2.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.4	22.4	16.9	12.5	21.1	15.8	11.6	20.7	59.6	26.7	55.5	25.4	52.0	23.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.3	3.6	3.2	-50.6	32.8	2.1	1.8	-52.1	34.1	-53.4	35.1	-54.5	36.1	-56.0	37.4		
			-50.6	21.8	2.1	1.1	-52.1	22.5	-53.4	23.2	-54.5	23.8	-56.0	24.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-7.0	-34.0	-28.1	-16.4	-3.9	6.1	14.2	16.9	14.3	5.5	-6.0	-23.1	-31.3
	DBStd	19.18	8.69	7.66	7.49	5.96	4.21	3.77	3.60	3.67	4.06	6.32	8.16	8.74
	HDD10.0	6723	1363	1067	817	417	132	8	1	7	142	497	993	1279
	HDD18.3	9296	1621	1301	1076	667	378	131	71	132	384	755	1243	1538
	CDD10.0	513	0	0	0	0	13	135	216	142	8	0	0	0
	CDD18.3	44	0	0	0	0	0	8	28	8	0	0	0	0
	CDH23.3	888	0	0	0	0	11	237	458	182	1	0	0	0
	CDH26.7	259	0	0	0	0	1	56	154	48	0	0	0	0
Wind	WSAvg	1.2	0.7	1.0	1.2	1.6	1.7	1.5	1.2	1.2	1.2	1.2	1.0	0.8
Precipitation	PrecAvg	435	18	12	13	16	36	63	72	68	58	32	22	21
	PrecMax	587	33	24	35	39	78	127	142	148	121	77	42	41
	PrecMin	290	8	5	2	0	5	11	17	16	17	9	6	7
	PrecStd	75	6	4	6	10	18	26	29	36	26	16	7	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-11.4	-5.6	6.7	15.0	25.3	30.8	33.1	31.1	21.7	10.7	4.2	-4.3
		MCWB	-12.2	-6.6	1.4	7.1	12.9	17.1	19.9	19.2	13.4	5.8	1.3	-5.5
	2%	DB	-14.4	-8.4	2.1	11.5	21.8	28.3	30.8	27.6	18.5	6.7	-3.6	-8.5
		MCWB	-14.9	-9.2	-1.1	4.4	11.1	16.2	19.1	18.5	11.6	3.7	-4.7	-9.0
	5%	DB	-16.9	-12.6	-0.3	9.0	18.6	26.5	28.2	25.1	15.9	4.1	-8.3	-13.3
		MCWB	-17.2	-13.3	-2.5	3.0	9.5	15.6	18.6	17.0	10.4	2.0	-9.1	-13.8
	10%	DB	-20.6	-15.6	-2.8	6.5	16.0	24.4	25.9	22.8	13.2	2.0	-11.3	-18.1
		MCWB	-20.8	-16.2	-4.5	1.7	8.4	15.0	17.8	16.3	9.2	0.2	-11.9	-18.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-12.1	-6.6	2.0	6.9	13.5	19.6	21.7	21.0	14.5	6.1	1.2	-5.5
		MCDB	-11.6	-5.7	6.1	14.0	23.6	25.3	27.9	26.9	20.2	9.6	4.0	-4.2
	2%	WB	-14.9	-9.2	-0.8	4.6	11.9	18.1	20.6	19.5	12.7	4.3	-4.5	-9.0
		MCDB	-14.5	-8.3	1.6	10.3	20.1	24.8	26.9	25.7	16.7	6.4	-3.7	-8.4
	5%	WB	-17.2	-13.3	-2.5	3.2	10.5	17.0	19.7	18.2	11.2	2.0	-9.1	-13.8
		MCDB	-16.8	-12.6	-0.5	8.3	17.0	23.8	25.8	23.4	14.6	3.5	-8.4	-13.4
	10%	WB	-20.9	-16.1	-4.5	1.8	8.9	15.9	18.7	17.1	9.7	0.5	-11.9	-18.4
		MCDB	-20.7	-15.6	-2.6	5.9	15.3	22.4	24.8	22.0	12.8	1.9	-11.3	-18.1
Mean Daily Temperature Range	5% DB	MDBR	9.6	14.5	18.8	15.4	13.4	14.8	13.6	14.2	11.9	9.6	10.8	9.2
		MCDBR	10.7	16.1	17.4	17.6	19.8	20.4	19.8	19.2	18.4	10.8	12.1	12.4
		MCWBR	10.2	15.3	14.4	10.8	10.8	9.5	9.3	10.3	11.6	7.6	11.1	12.1
	5% WB	MCDBR	10.6	15.8	16.1	15.7	17.2	16.8	15.4	15.9	15.2	9.8	11.7	12.4
		MCWBR	10.2	15.1	13.5	9.7	9.7	8.6	8.0	8.8	10.1	7.1	10.8	12.0
Clear-Sky Solar Irradiance	taub	0.249	0.283	0.317	0.390	0.466	0.437	0.494	0.438	0.350	0.305	0.255	0.230	
	taud	1.986	2.018	2.015	1.928	1.978	2.208	2.054	2.200	2.403	2.245	2.013	1.928	
	Ebn at Noon	533	711	802	796	760	798	734	752	781	692	513	408	
	Edh at Noon	53	87	119	156	161	130	149	120	83	69	50	39	
All-Sky Solar Radiation	RadAvg	0.62	1.68	3.44	5.19	5.02	5.57	5.06	3.96	2.53	1.34	0.79	0.38	
	RadStd	0.04	0.08	0.13	0.22	0.41	0.43	0.44	0.30	0.19	0.13	0.07	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	+0.84	N/A	N/A	N/A	N/A

Nomenclature: See separate page